

“The idea is pretty simple. We take the Landsat satellite archive data, stack the satellite imagery on top of each other, and look from top to down at each pixel. We pick up the frequency of the presence or absence of water. From the pixels we can decide whether it’s a permanent or temporary water body. People have used this before, but for short periods of time. It has never been released as a tool that can be used globally,” said Dr Mehmood.

Flood maps help communities to know which areas have the highest risk of flooding. “What we want to do is to take risk-mapping down at the street level. We can’t say the whole district is high risk; that doesn’t make sense,” he added.

To be open sourced soon

This tool will soon be open sourced. “We will be publishing the tool on INWEH’s GitHub account under the Academic Free License v3.0. The details for the algorithm are already published as an open-access publication,” said Dr Mehmood.

Dr Mehmood, who traces his roots to South Asia and led the tool’s development, also stated that a UNU-INWEH survey showed a majority of flood forecasting centres in flood-prone countries “lack the ability to run complex flood forecasting models.”

Using Landsat and the Google Earth Engine Cloud

The tool offers the mapping of flood areas, using Landsat with the Google Earth Engine Cloud platform. It has deployed Landsat data since 1985. Landsat is the longest-running enterprise for collecting satellite imagery of Earth, run by NASA and the US Geological Survey.

Layers of Landsat information for a selected region and specified timeframe identify temporary and permanent water bodies while integrating site-specific elevation and land-use data. This produces a detailed map of flood inundation in recent decades, with

available overlays of population, buildings and land use, which can be used for community planning, as well as building zoning and insurance assessments. To validate the technology, maps generated in less than a minute using the new tool were compared to documented flooding events in Australia, Bangladesh, Canada, Cambodia, India, Mozambique, Sri Lanka and Thailand, with 82 per cent accuracy achieved, the UNU said.

Partners for this project include tech giant Google (offering in-kind support through the Google Cloud for Researchers programme); MapBox, the San Francisco-based provider of custom online maps for websites and applications; and India’s International Institute of Information Technology, known for its narrow and deep focus on computer science.

This tool is promoted as being “simple to use.” All one needs is Internet access to visit floodmapping.inweh.unu.edu and obtain a flood map at 30-metre resolution -- street by street level.

Future plans

More solutions are planned after the Flood Mapping Tool, or FMT. A Web based Spatial Decision Support System

(WSDSS) aims to address information gaps in flood early warning and risk management systems. The WSDSS will include FMT and a flood risk prediction tool (FRPT).

The FRPT is due to be released by 2022, using AI models to generate current and future flood risk maps for three climate change scenarios — at the city, district, and river basin levels. The AI models will be trained using the inundation maps generated by the FMT and open data sets, including land use, land cover, precipitation, temperature, gender, and age-disaggregated socio-economic data.

The WSDSS will improve the spatial and temporal coverage of early warning and risk management. It hopes to build the capacity of flood forecasting centres in the ‘Global South’ to use AI models, Big Data and cloud computing to analyse the impacts of climate change.

The Flood Mapping Tool has been developed over two years. Technical and national disaster management experts were engaged from Afghanistan, Bangladesh, Bhutan, India, Pakistan, Nepal, and Sri Lanka to design, review and test the tool.



Hyderabad 2020 floods. Photo courtesy: Wikimedia Commons/ Strike Eagle